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AI-BASED AUTOMATED CLEARANCE SYSTEM

¹J MUNI PARTHA, ²CHANDU TANGELLA

^{1,2}Vishnu Institute of Technology, Bhimavaram, Andhra Pradesh, India.

¹munipartha.j@vishnu.edu.in, ²chandutangella9@gmail.com.

Abstract— The AI-Based Automated Clearance System is designed to modernize and optimize the conventional university clearance procedure, which typically requires students to obtain signatures and approvals from multiple departments before receiving academic documents. The proposed system automates the entire workflow beginning from student registration, secure login, and submission of required documents such as library dues, laboratory records, hostel information, and administrative forms. Each department accesses a unified platform to review student submissions, validate compliance with institutional requirements, and update approval statuses. The system integrates automated decision-support mechanisms that detect missing documents, highlight incomplete verifications, and notify students instantly, reducing repetitive manual communication. Department-wise verification is performed sequentially or in parallel based on institutional policies, and the system ensures transparency by allowing students to track their real-time clearance progress at every stage. Once all departments have completed their verification, the system automatically generates the final clearance approval, which students can download digitally. By eliminating paperwork, minimizing human error, and shortening processing time, this AI-enabled framework significantly enhances operational efficiency and ensures a more reliable, accessible, and user-friendly clearance experience for both students and administrative staff.

Keywords— Digital clearance management, AI-driven verification, transformer-based automation, No Due certificate generation, departmental fee integration, secure academic workflow.

I. INTRODUCTION

Clearance is a formal certification granted to individuals, including university students, government employees, and military personnel, to authorize access to sensitive information or confirm the completion of obligations before disengagement [1]. In higher education institutions, students must obtain clearance from multiple departments such as library, hostel, accounts, and academic offices to receive essential credentials, including transcripts, degree certificates, and transfer documents [2]. Traditionally, this process has relied on manual verification, physical documentation, and in-person approvals, resulting in delays, administrative bottlenecks, and a higher likelihood of human error [3]. The rapid development of digital technologies has enabled efficient record management, allowing institutions to store and retrieve information electronically while reducing manual effort and improving accuracy [4].

Despite the transition to digital systems, existing clearance management solutions exhibit notable limitations. Many systems still require manual intervention for verification, causing delays and inconsistencies in processing [5]. Coordination across departments is often inadequate, limiting real-time tracking of obligations and hindering timely issuance of clearance documents [6]. Furthermore, the lack of secure digital certification and automated notifications reduces transparency, increases administrative workload, and

diminishes student convenience [7]. These gaps underscore the need for a centralized, intelligent framework capable of managing multi-departmental clearance efficiently and reliably [8].

The objectives of this study are to design a centralized clearance management platform that automates verification, consolidates departmental records, and generates digitally certified clearance documents while notifying students and administrators in real time [9]. The system ensures accurate monitoring of fee payments, departmental approvals, and completion of obligations, thereby reducing reliance on manual processes and enhancing operational efficiency. By providing a unified interface for all stakeholders, the framework improves accountability, consistency, and transparency across the entire clearance workflow.

The significance of the proposed framework lies in its potential to enhance institutional efficiency, reduce administrative workload, and improve student satisfaction [10]. Automated clearance management streamlines operations, minimizes errors, and provides secure, verifiable documentation. Real-time tracking and centralized monitoring allow administrators to oversee departmental processes effectively, while students benefit from convenient access to clearance status and certified documents. This approach establishes a scalable, reliable, and impactful solution for modernizing academic administration.

II. RELATED WORK

Several studies have explored digital clearance and academic management systems to improve efficiency and reduce manual administrative burdens. Nneji et al. [11] developed an improved e-clearance management system for graduating students, focusing on streamlining the verification of student obligations across multiple departments. Their system demonstrated faster clearance processing and reduced paperwork, but relied heavily on manual input and lacked real-time automated verification across departments. AISideiri et al. [12] proposed a web-based online clearance system that facilitated fee payment tracking and document verification, enabling students and staff to access clearance status remotely. While this system improved accessibility, it had limited integration with departmental databases and minimal intelligent decision-making capabilities.

Rana et al. [13] designed an automated academic assignment submission and management system using agile methodology. The system emphasized centralized tracking and accountability of student submissions, highlighting the benefits of digital process management. However, its application was limited to assignment handling and did not extend to comprehensive clearance procedures. Seezi et al. [14] developed an online clearance system for higher education institutions, focusing on digitizing the entire clearance workflow. Despite improvements in transparency

and process monitoring, the system lacked secure digital certification and automated notification mechanisms.

Medak and Gogoi [15] explored university admission management systems, demonstrating effective student data handling, enrollment verification, and administrative oversight. While efficient for admissions, the system did not address multi-departmental clearance or cross-verification of dues. Ahmed et al. [16] proposed a web-based student registration and exam form management system, enabling seamless student interactions with administrative modules. Its limitations included the absence of automated clearance validation and lack of real-time monitoring of pending obligations. Wahab et al. [17] developed a web-based student portal offering centralized access to academic records and administrative updates. Although the system enhanced user convenience and engagement, it did not incorporate intelligent verification or cross-department automation.

IFEANYI [18] implemented an online clearance system emphasizing workflow digitization and reduction of manual interventions. While it provided a foundational framework for automated clearance, the system lacked advanced decision-making capabilities and secure digital certification. Collectively, these studies demonstrate the benefits of digitalization in reducing manual effort, enhancing accessibility, and improving administrative tracking. Nonetheless, challenges such as real-time verification, intelligent cross-departmental integration, secure certificate generation, and automated notifications remain unaddressed.

The current study addresses these gaps by developing a centralized, intelligent clearance management platform that integrates multiple departmental modules, verifies dues automatically, generates digitally certified No Due Certificates, and provides timely notifications to students and administrators. By combining comprehensive record consolidation, real-time verification, and secure document issuance, the system ensures accuracy, transparency, and efficiency, overcoming limitations observed in prior research. This approach establishes a scalable and reliable framework capable of modernizing clearance management across higher education institutions.

III. MATERIALS AND METHODS

The AI-Based Automated Clearance System provides a centralized digital platform that automates the complete student clearance workflow across all departments. After student registration and secure login, the system enables students to upload required documents and view departmental requirements. Each department—such as library, hostel, accounts, and academic office—accesses its respective module to verify dues, update approval status, and record pending obligations. The system employs AI-driven decision-support logic to identify missing documents, flag incomplete submissions, and generate automated alerts to both staff and students. A role-based access framework ensures that administrators can manage user accounts, departments can validate records and update clearances, and students can track their real-time progress through a unified dashboard. Status tracking modules allow students to monitor the verification stages and receive automated notifications at each step. Once all departmental verifications are completed, the system automatically consolidates approvals and generates the final digital clearance certificate, which includes secure

authentication features and is accessible for student download. This workflow-oriented architecture enhances transparency, reduces manual errors, minimizes administrative delays, and ensures fast, reliable, and streamlined clearance processing.

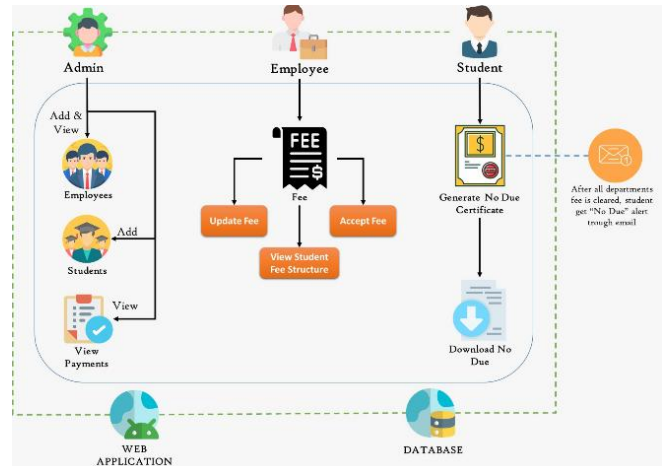


Fig. 1. System Architecture

The system architecture illustrates a centralized AI-based clearance platform integrating multiple user roles and departmental modules. Administrators manage student and employee records, while employees update and accept fees, and students monitor their dues. Key operations include updating fees, accepting payments, and viewing student fee structures. The architecture incorporates a secure database for record storage and retrieval, ensures automated verification, and facilitates generation of digitally signed No Due Certificates. Notifications and document downloads provide real-time communication, promoting efficiency, transparency, and seamless coordination across all departments.

A) Modules:

i) Admin Module:

Admin Login: The administrator accesses the system through a secure login interface that validates username and password credentials. Upon successful authentication, the system grants access to the administrative dashboard. Invalid credentials trigger an error message and deny entry to ensure secure and restricted access.

Add Employee: The administrator can register new departmental employees by entering details such as personal information, qualifications, experience, contact details, and assigned department roles. The system validates the uniqueness of login credentials before storing the record. Successful entries generate a confirmation message, while duplicate usernames result in an error notification.

View Employees: The system retrieves and displays a complete list of all registered employees. Each record includes detailed information along with management options. The administrator can modify or remove employee profiles directly from this interface, enabling efficient personnel management across departmental units.

Update Employee: When the administrator selects an employee for modification, the system loads the existing data for editing. Upon submission, the updated information is

validated and stored in the database. A confirmation message is displayed after successful updates, ensuring the accuracy and maintenance of employee records.

Delete Employee: The administrator may delete an existing employee profile when required. Upon confirmation, the system removes the employee record from the database and updates the list to reflect the change. This ensures that only active and authorized personnel remain in the clearance system.

Add Student: New student profiles can be created by providing personal details, academic information, and fee components associated with various departments. After validation, the student record is inserted into the system database. The administrator receives a success or error message depending on data validity and duplication checks.

View Students: The system provides a complete list of student records, including academic details and department-wise fee information. The administrator can view, modify, or delete any student entry, enabling effective oversight of student data relevant to the clearance process.

Update Student: When editing is required, the system retrieves the existing student information for modification. After the administrator submits the updated details, the changes are saved to the database and confirmed with a notification. This ensures that all student data remains current and accurate.

Delete Student: A student profile can be removed following administrator confirmation. Once approved, the corresponding record is deleted from the system, and an updated list is displayed. This operation ensures the maintenance of valid and active student datasets.

View Payments: The administrator can access consolidated payment information for any student by selecting the relevant identifier. The system retrieves department-wise payment history, showing paid amounts, dues, clearance status, and transaction details. This unified payment view supports monitoring of financial compliance and clearance progression across all departments.

ii) Employee Module:

Employee Login: Employees authenticate using their assigned credentials and selected departmental role. The system validates the submitted username, password, and department against registered employee records. Successful authentication initializes session variables that store the employee's identity and department, enabling department-restricted operations. Failed authentication results in an error message, ensuring controlled and secure access to all clearance-related functions.

Accept Fee: Employees can record departmental fee payments made by students. The interface allows selection of a student, entry of fee amount, and specification of academic year. The system performs validations—including numeric checks, course-year consistency, and verification of departmental fee structure—before inserting the transaction into the centralized payment database. Successful submission updates payment logs in real time, contributing to accurate due calculation for automated clearance processing.

Update Fee: Employees may adjust the department-specific fee assigned to a student for the relevant academic year. Upon selection of the student and entry of the revised amount, the system validates all inputs and maps the update to the correct fee component based on the employee's department (tuition, library, hostel, exam, or laboratory). Valid updates modify the corresponding student record in the database and immediately reflect in AI-based due verification processes used for clearance certification.

View Payments: Employees can retrieve and review detailed payment histories for any student within their department. The system filters records based on the employee's departmental assignment and displays date-wise payments. For each transaction, the system computes the total fee assigned, cumulative amount paid, and remaining due. Clearance status is dynamically determined from these calculations, enabling employees to monitor financial compliance efficiently and support accurate, real-time clearance generation.

View Students: Employees have read-only access to the complete student list stored in the system. The module presents academic and fee-related information without allowing edits, providing employees necessary visibility for decision making in payment and verification tasks while preserving data integrity.

iii) Student Module:

Student Login: Students authenticate using their unique ID and password. The system validates these credentials against registered student records and, upon successful verification, initializes the session using the student's identity. All academic-year records associated with the student are retrieved, and personalized details are loaded into the dashboard. Failed authentication triggers an error message, ensuring secure access to the clearance system.

Generate No Due Certificate: When students select an academic year, the system retrieves departmental fees and corresponding payments, evaluating each category as cleared or pending. It then generates a structured fee summary with certificate metadata, including a verification hash. If all dues are settled, the certificate becomes downloadable, status flags are updated, and a confirmation email is sent. If dues remain, a color-coded pending-fee breakdown is shown.

Download Certificate: Students who have achieved full clearance may download an authenticated no-due certificate. The system can generate a PDF certificate, a QR-code validation file, or both as a bundled archive. The PDF includes institutional branding, certificate metadata, fee tables, clearance status indicators, department signatures, and a unique cryptographic certificate number. The QR version encodes student details, certificate number, issuance date, and verification status. The system returns the requested file format directly for secure download.

Email Notification: When a student successfully clears all departmental dues, the system automatically retrieves the student's registered email address and sends a confirmation message. The notification informs the student that the certificate has been generated and is available for download.

ensuring timely communication and improved user engagement.

iv) Public/QR Code Verification:

QR Code Scanning Interface: The system provides a public QR-scanning interface that allows users to authenticate certificates without requiring login credentials. When the QR code printed on a certificate is scanned, the encoded student ID and academic year are extracted. This scanning process acts as the first validation layer, enabling fast and secure access to certificate verification services.

Certificate Verification Service: Once the QR code is scanned, the system uses the extracted student ID and academic year to verify record availability and retrieve departmental fee data. It compares assigned and paid fees to determine clearance status. If all dues are cleared, the system returns validated student and certificate details; otherwise, it reports pending fees and marks the certificate as invalid.

Certificate Retrieval Through QR Validation: After successful validation, users can download the authenticated certificate through the QR workflow. The system collects student information and fee summaries, formats them into a structured dataset, and generates an official PDF identical to the originally issued version. The file includes fee details, clearance status, digital signatures, issuance date, and a unique hash, ensuring secure, verified access to clearance documents.

B) Methods/Techniques:

Rule-Based Verification Logic: The system uses structured rule-based logic to evaluate fee clearance for each department. For every certificate request, the system compares the total assigned fee with cumulative payments recorded in the database for that specific year. A fee is marked as cleared only when paid amounts meet or exceed departmental totals, while any shortfall automatically triggers a pending status. This deterministic verification ensures transparency, prevents incorrect clearances, and guarantees consistent processing across all workflows.

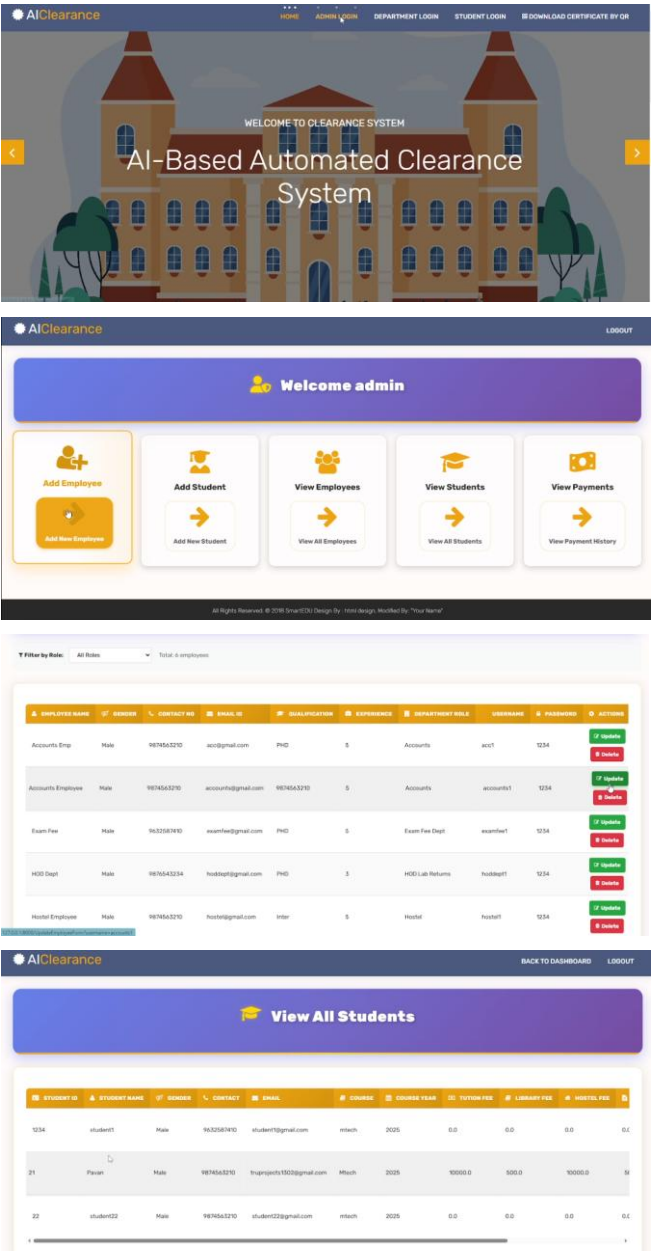
QR-Code-Based Certificate Validation: A QR-based validation mechanism enables instant verification of generated certificates. Each QR code encodes the student ID, academic year, certificate number, and verification parameters. When scanned, the system retrieves fee and payment details and re-evaluates clearance status in real time. Only if all dues are cleared does the system confirm certificate validity. This method prevents forgery, supports offline verification through encoded data, and strengthens document authenticity.

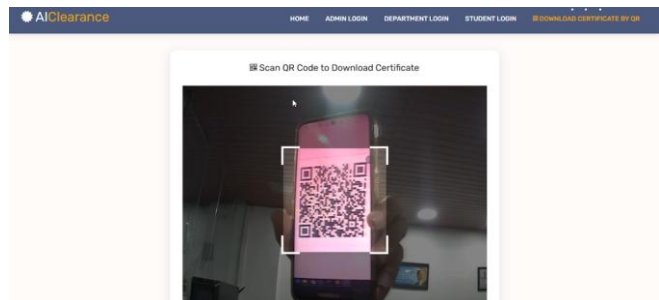
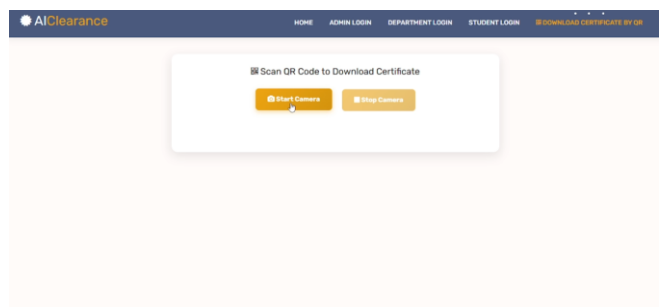
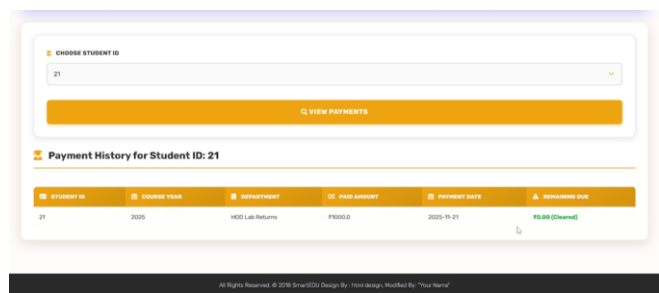
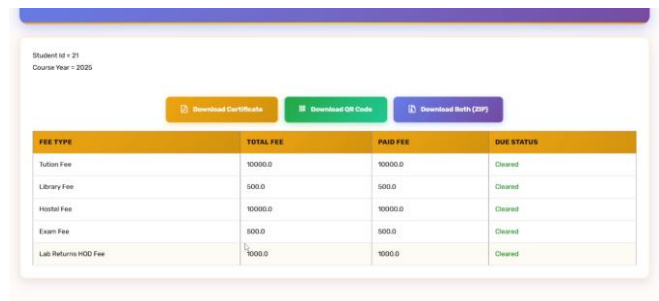
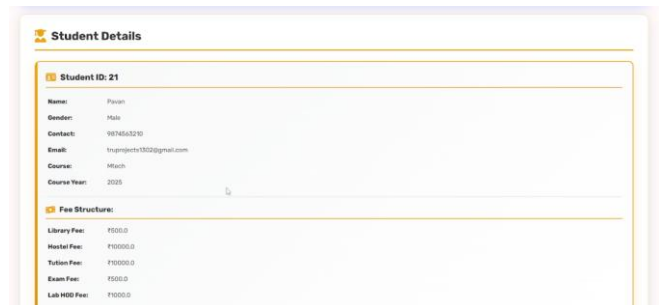
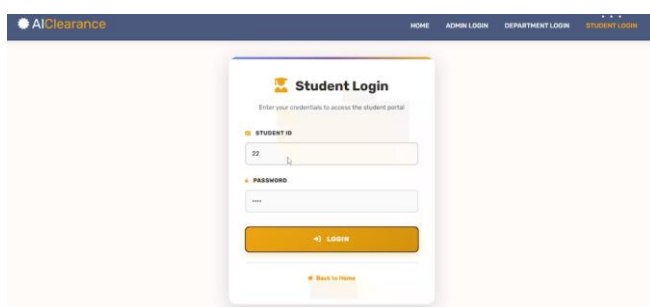
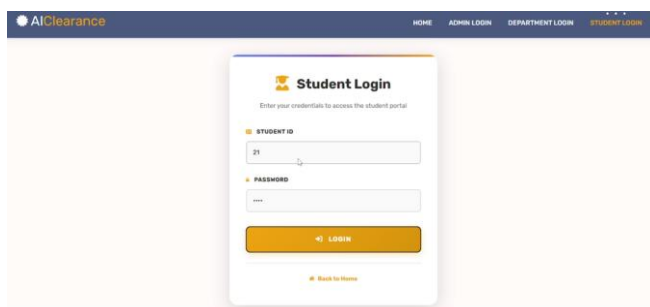
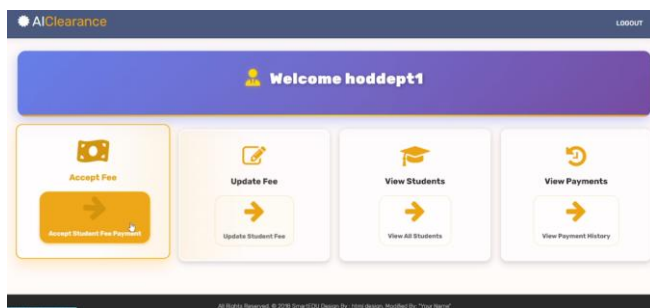
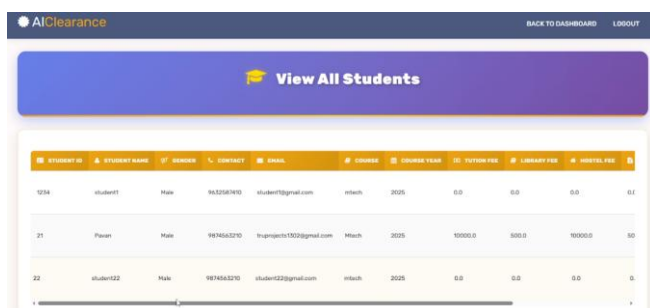
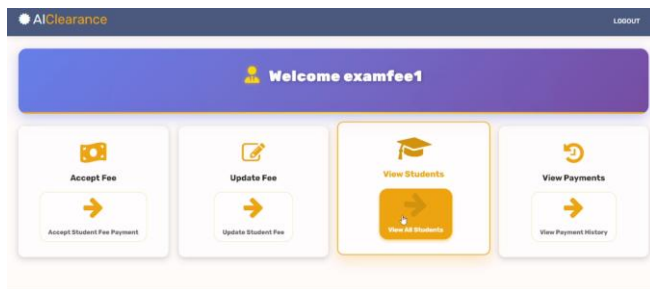
Digital Certificate Generation (PDF with Structured Data): Certificate creation is performed using automated PDF generation techniques that embed verified student details, fee breakdowns, color-coded clearance status, certificate hash, issue date, and department signatures. The system constructs a fee data table from real-time calculations and formats it into a standardized certificate layout. This approach ensures consistent, tamper-resistant, and institution-approved documentation for academic and administrative use.

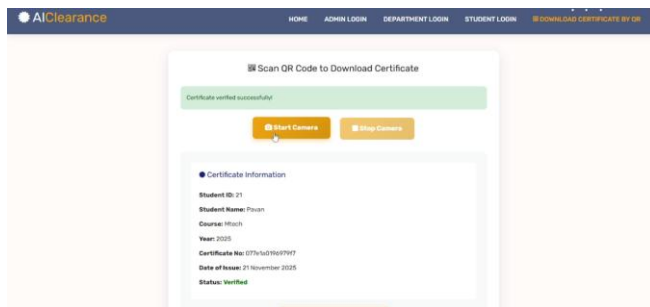
Database-Centric Operations (CRUD + Aggregation Queries): The workflow relies heavily on relational database operations for managing employees, students, fee structures, and payment histories. CRUD operations support registration, updates, and deletions, while aggregation queries compute cumulative payments and remaining dues for each department. This structured data handling ensures accuracy in clearance evaluation, real-time record updates, and seamless coordination between admin, employee, and student modules.

Automated Email Notification (SMTP): An SMTP-based notification system automatically sends emails to students once their dues are verified and a No Due Certificate is successfully generated. The system retrieves the student's email address, composes a verification message, and dispatches it through a secure mail server. This automation reduces manual communication efforts, provides timely updates, and keeps students informed about certificate availability.

IV. EXPERIMENTAL RESULTS







V. CONCLUSION

In conclusion, the AI-Based Automated Clearance System provides a comprehensive and streamlined digital framework that transforms the traditionally manual and time-consuming student clearance process into an efficient, transparent, and unified workflow. By integrating multiple departmental modules into a centralized platform, the system enables secure role-based access for administrators, staff, and students, ensuring that each stakeholder can perform their responsibilities without dependency on physical coordination. Departments update dues, approvals, and clearance statuses directly within the system, allowing real-time synchronization of information and eliminating delays caused by repeated manual verification. Students can continuously track their progress through an intuitive dashboard, greatly improving clarity and reducing uncertainty during the clearance period. Once all departmental validations are completed, the system automatically generates a digitally authenticated clearance certificate and sends notifications through secure channels to ensure seamless communication. The automation of data verification, centralized record management, and structured workflow execution collectively reduce administrative workload, enhance accuracy, and accelerate completion timelines. Overall, the system delivers a scalable, secure, and user-centric solution that strengthens institutional efficiency and provides a more convenient and reliable clearance experience for students and staff.

The AI-Based Automated Clearance System can be further enhanced by integrating advanced machine learning algorithms to predict potential clearance delays and recommend proactive actions. Mobile applications can be developed to provide students with real-time updates and notifications on the go. Integration with biometric authentication and blockchain technology can ensure higher security, immutability, and authenticity of clearance certificates. Expanding the system to accommodate additional institutional processes, such as internship approvals, exam registrations, and alumni tracking, can increase its utility. Furthermore, cloud-based deployment can improve scalability, accessibility, and cross-campus synchronization, making the system adaptable for large universities and multi-campus environments.

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